

Once installation is complete, we need to make some changes in Samba's configuration file. But before that, create back up of the configuration file (in case we make any mistake) by executing the command below as shown in Fig. 6:

```
$ sudo cp /etc/samba/smb.conf /etc/samba/smb.conf.old
```

This will create a backup of the configuration file with the name 'smb.conf.old' in the same directory.

Now to edit the Samba config file, open it in nano editor by executing the command below as shown in Fig. 7:

```
$ sudo nano /etc/samba/smb.conf
```

The first entry in this file is workgroup identifier; by default it is 'workgroup = WORKGROUP.' If you are using a different name for your home workgroup, go ahead and change that now; otherwise leave it as the default.



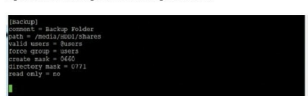
```
pi@raspberrypi:~$ sudo cp /etc/samba/smb.conf /etc/samba/smb.conf.old
```

Fig. 6: Backup of the configuration file



```
pi@raspberrypi:~$ sudo nano /etc/samba/smb.conf
```

Fig. 7: Edit the configuration file using nano editor



```
(Backup)
comment = Backup Folder
path = /media/HDD1/shares
valid users = @users
force group = users
create mask = 0660
directory mask = 0771
read only = no
```

Fig. 8: Adding the lines to configuration file



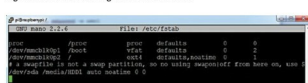
```
pi@raspberrypi:~$ sudo /etc/init.d/samba restart
```

Fig. 9: Restart Samba



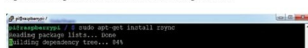
```
pi@raspberrypi:~$ sudo smbpasswd -s -u user1
New SMB password:
Retype new SMB password:
pi@raspberrypi:~$
```

Fig. 10: Add user1 as legitimate Samba user



```
File: /etc/fstab
#  proc      /proc      proc      defaults    0    0
/dev/mmcblk0p1 /boot      vfat      defaults    0    2
/dev/mmcblk0p2 /          ext4      defaults,noatime 0    1
# a swapfile is not a swap partition, so no using swapon/off
# From here on, use #
/dev/sda /media/HDD1 auto noatime 0 0
```

Fig. 11: Configuration for automatic mounting of drives



```
pi@raspberrypi:~$ sudo apt-get install rsync
Reading package lists... done
Building dependency tree... 0/4
```

Fig. 12: Installing the rsync



```
pi@raspberrypi:~$ crontab -e
```

Fig. 13: For automatic process of copying files

Under authentication section, add following line or remove the '#' symbol if it is already there. Otherwise, anyone with network access will be able to access the shared data. This will enable username and password verification.

```
security = user
```

Now scroll down to the bottom of the file and add following commands as shown in Fig. 8:

```
(Backup)
comment = Backup Folder
path = /media/HDD1/shares
valid users = @users
force group = users
create mask = 0660
directory mask = 0771
read only = no
```

After adding this, press 'CTRL+O' to save the file and 'CTRL+X' to exit.

Restart Samba as shown in Fig. 9 with the command below:

```
$ sudo /etc/init.d/samba restart
```

Add in a user who can access Raspberry Pi's Samba with the commands below. In this case, user is 'user1' and password is 'password123.' You can choose any username and password.

```
$ sudo useradd user1 -m -G users
$ sudo passwd password123
```

Reconfirm the password when prompted. After confirming the password, it is time to add 'user1' as a legitimate Samba user. Enter the following command:

```
sudo smbpasswd -s -u user1
```

Enter the password when prompted as shown in Fig. 10.

You can now use any Samba-capable machine on network and test connectivity to the network using

the username and password created earlier.

Configuring Pi for automatic mounting of the drives

We need to configure our Raspberry Pi such that when it restarts, it automatically mounts the external hard drives. To do so, we need to start the nano editor and make following changes:

```
sudo nano /etc/fstab
```

Add the following lines in the file as shown in Fig. 11:

```
/dev/sda1 /media/HDD1 auto noatime 0 0
/dev/sdb2 /media/HDD2 auto noatime 0 0
```

After adding these lines, press 'CTRL+O' to save the file and 'CTRL+X' to exit.

Configuring Raspberry Pi for data redundancy

To configure Raspberry Pi to perform data mirroring from the share folder of the primary disk to the share folder of the secondary disk, two tools have to be used—'cron' and 'rsync.'

Install rsync using the command:

```
$ sudo apt-get install rsync
```

Once rsync is installed, set up cron to automate the process of copying files from HDD1 to HDD2. Run the command:

```
$ crontab -e
```

This will open up cron scheduling table. Add following line at the end of the file:

```
0 5 * * * rsync -av --delete /media/HDD1/shares /media/HDD2/shares/
```

In section 0 5 * * *, first two digits define time and the three stars are for place of year, month and day as wild card entries.

Thus each day at 5:00 am, rsync will compare the share folders of the primary and secondary drives and copy all new files to secondary drive. It will also delete those from the secondary drive which are no longer present in the primary drive. Now save and exit the editor. ●

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